

ea适配调优文档

环境准备

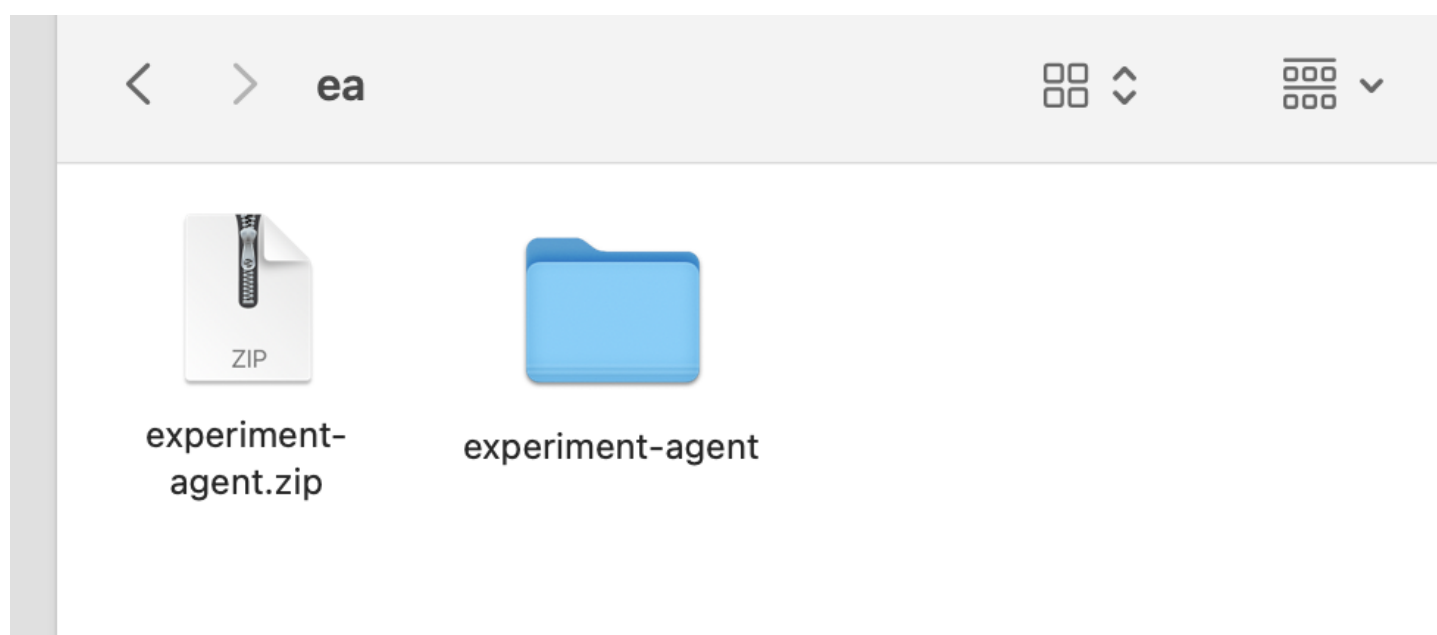
系统环境必须安装有conda与docker

源代码:

<https://github.com/hitsz-ids/AI-Course.git>

Agentic_AI_M2/ea调优/code/experiment-agent.zip

解压源代码到文件夹:

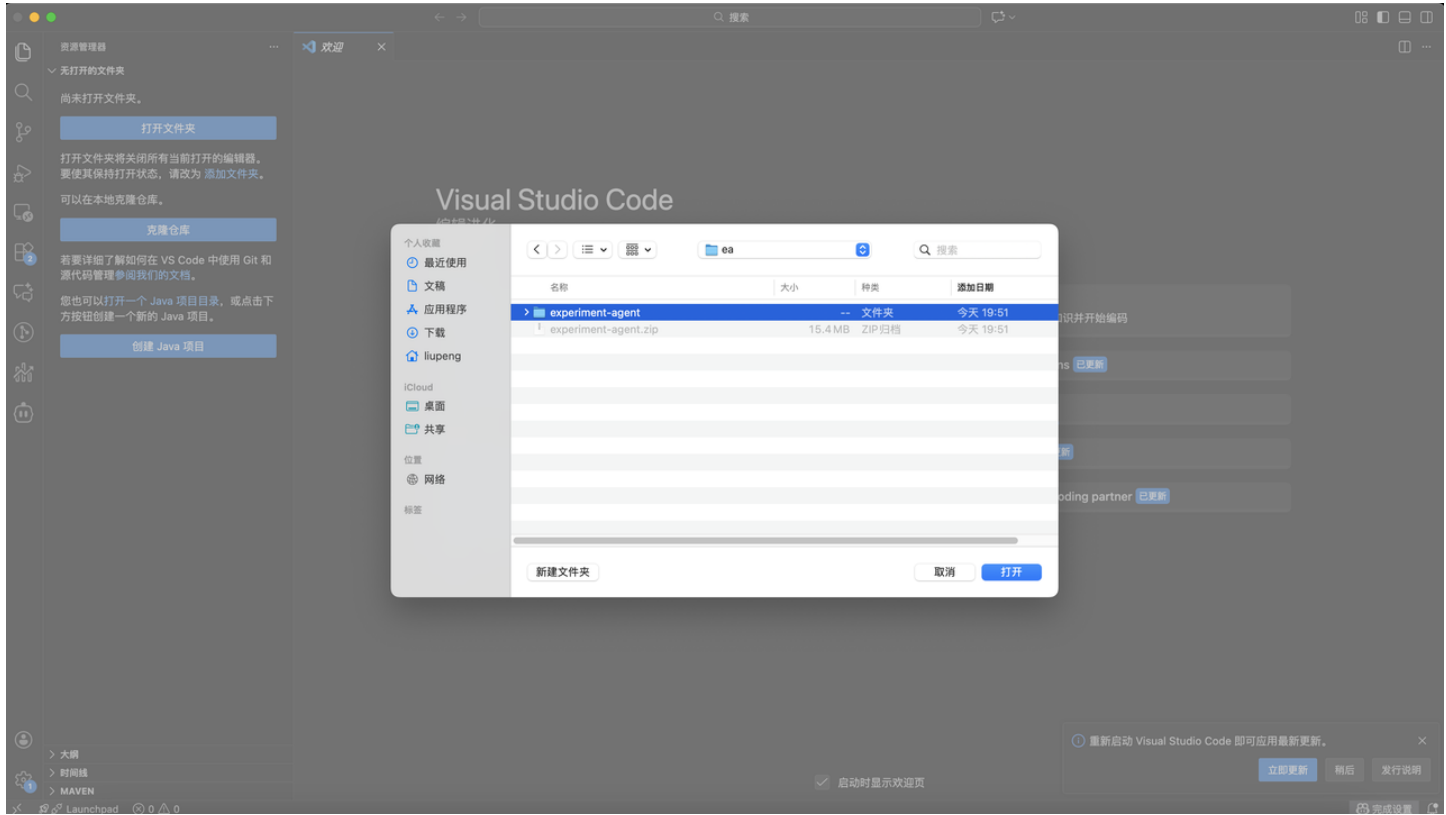


使用命令行解压:

代码块

- 1 #将压缩包放到文件夹解压
- 2 unzip experiment-agent.zip

打开VSCode选择已解压项目



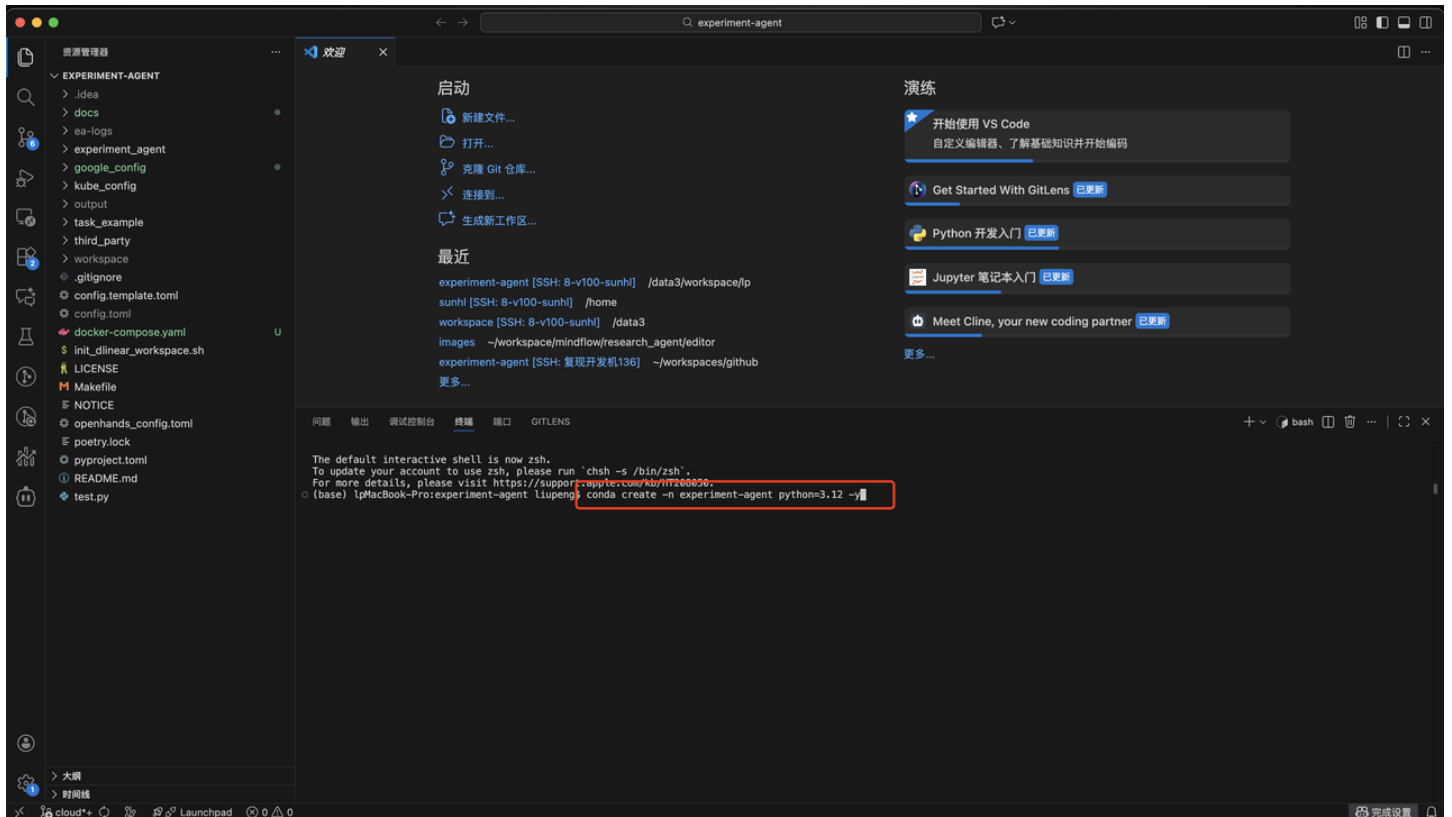
创建ea-python环境

1.新建conda环境

新建终端,并创建experiment-agent conda环境

代码块

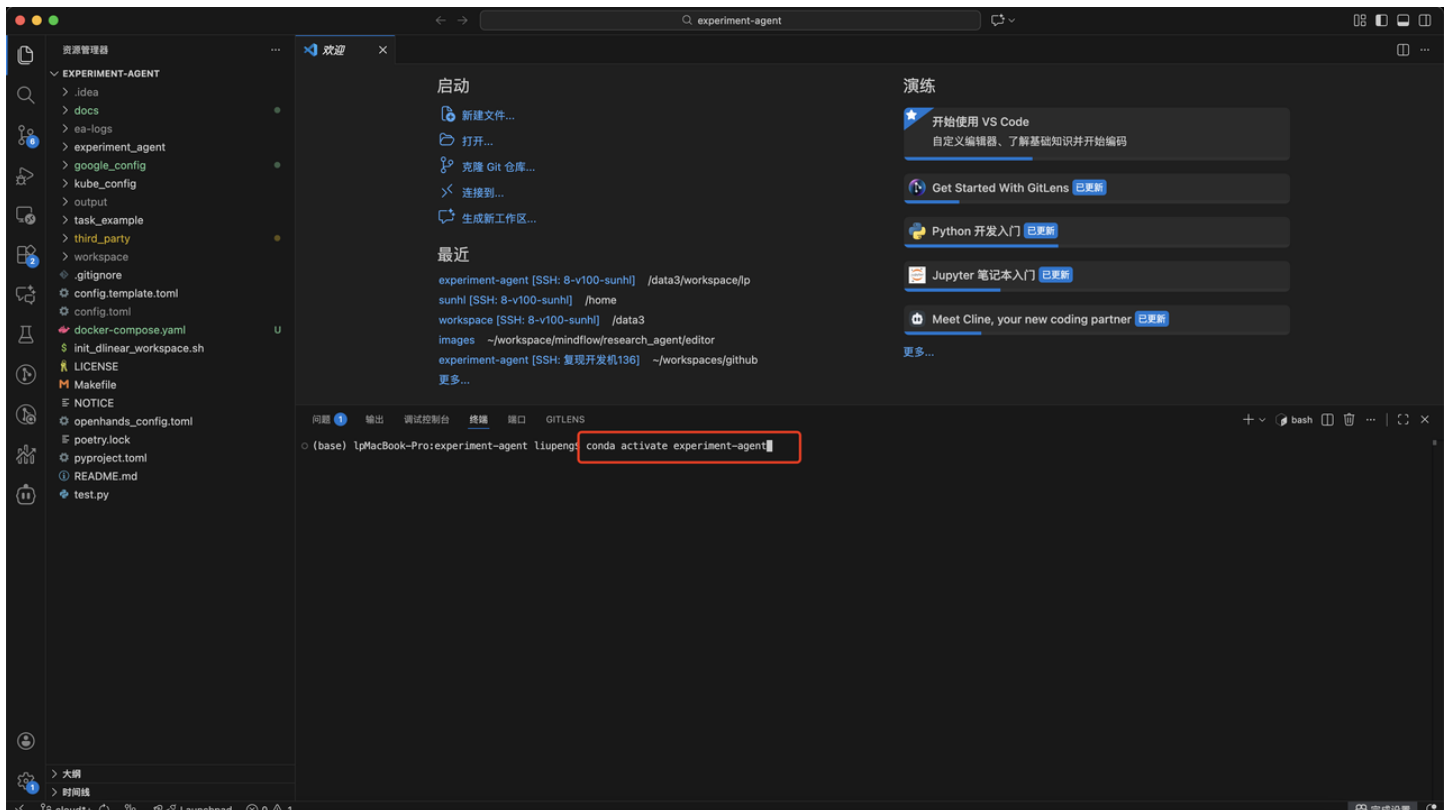
```
1 conda create -n experiment-agent python=3.12 -y
```



2. 激活conda环境

代码块

```
1 conda activate experiment-agent
```

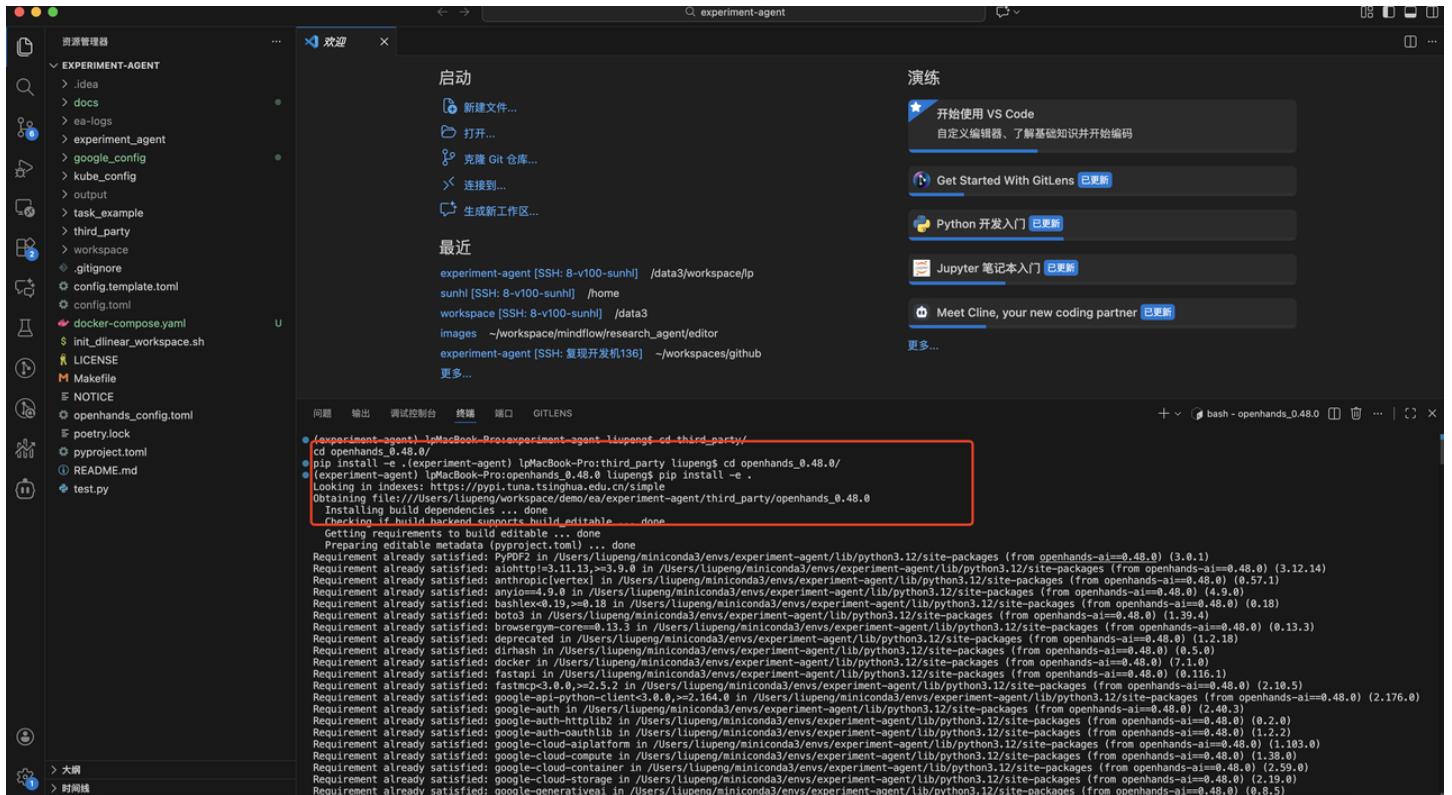


3. 依赖安装

3.1 openhands依赖安装

代码块

```
1 cd third_party/
2 cd openhands_0.48.0/
3 pip install -e .
```



完成依赖后终端指令退回到项目根目录

代码块

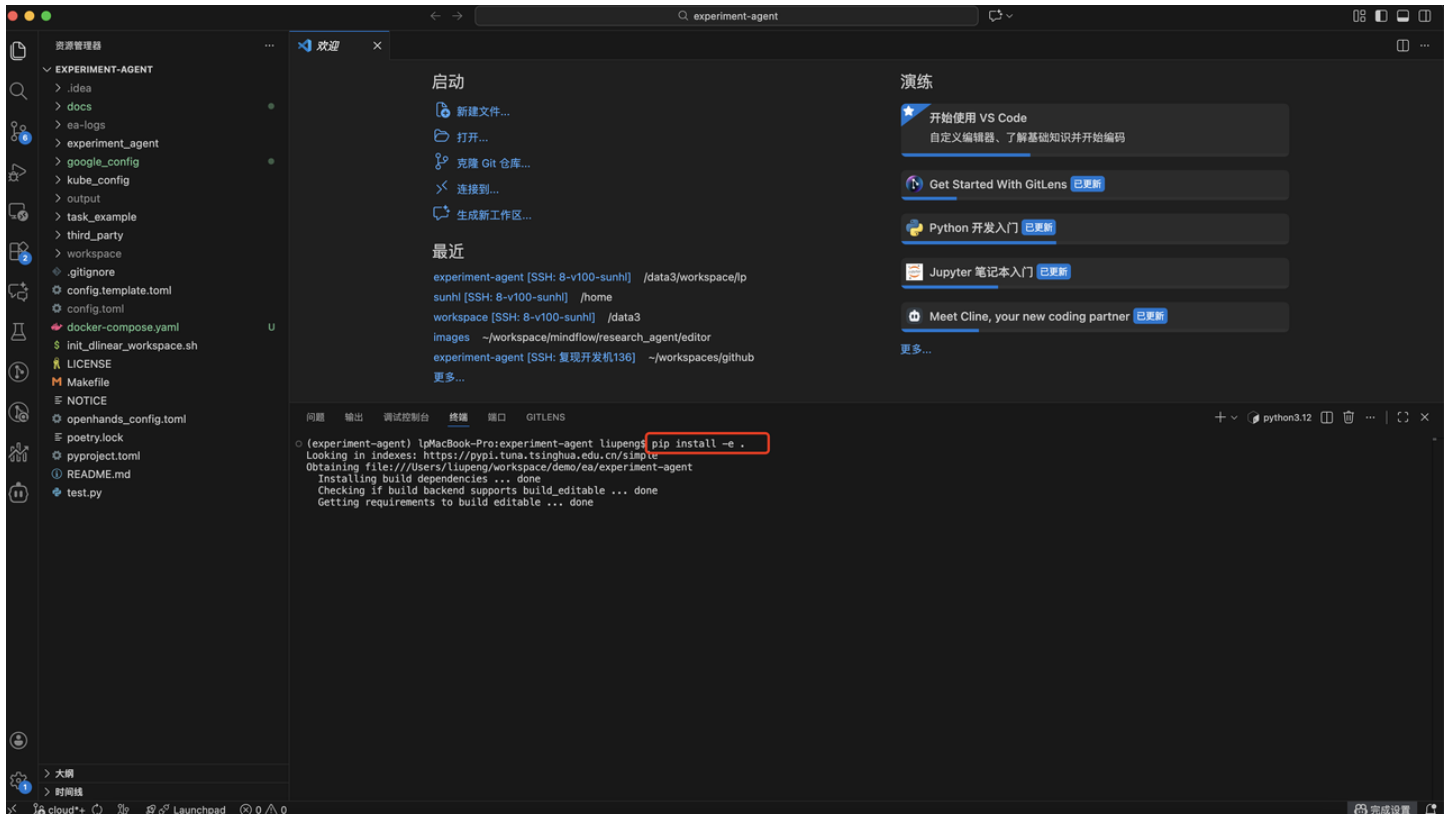
```
1 cd ../
2 cd ../
```

3.2 ea依赖安装

在ea项目根目录执行命令

代码块

```
1 pip install -e .
```



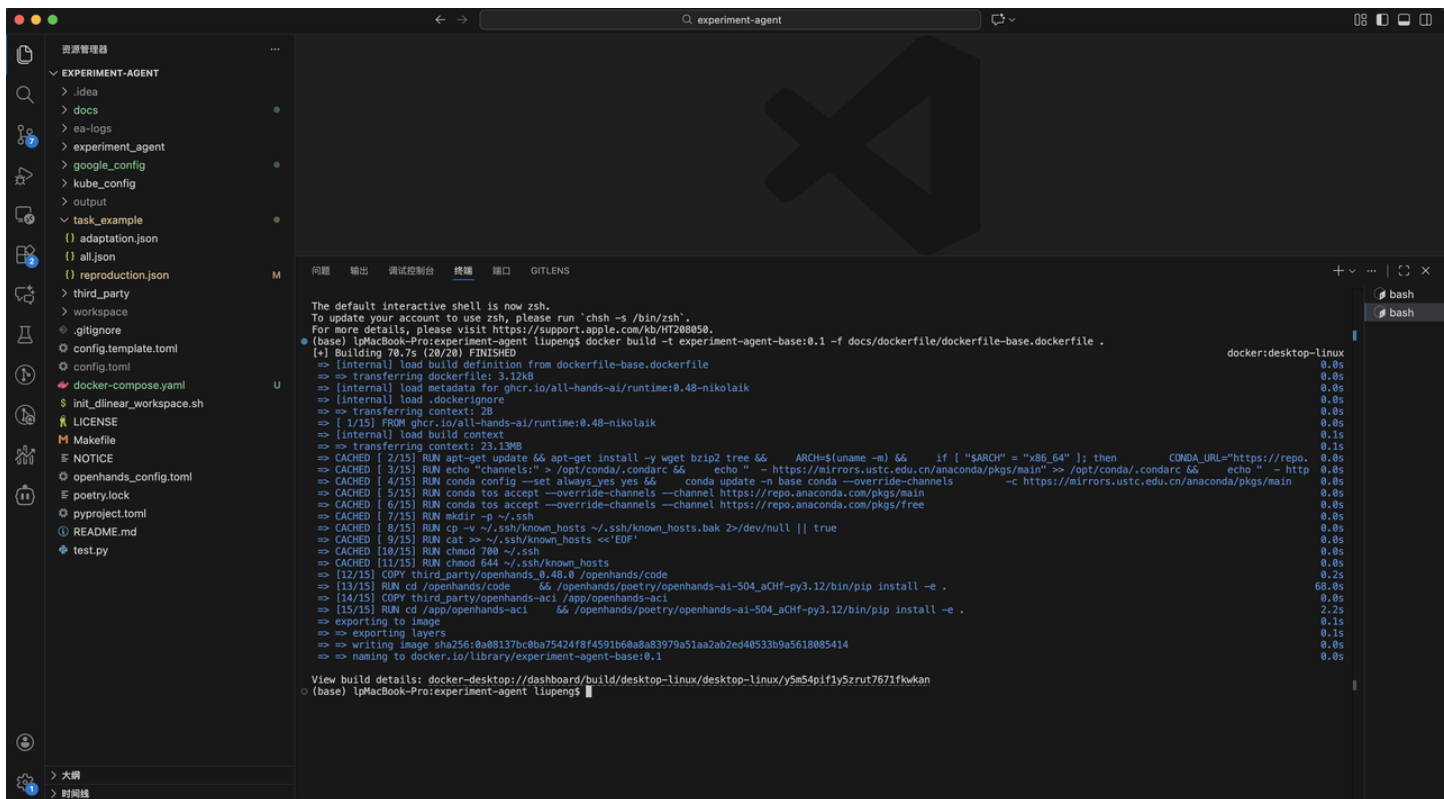
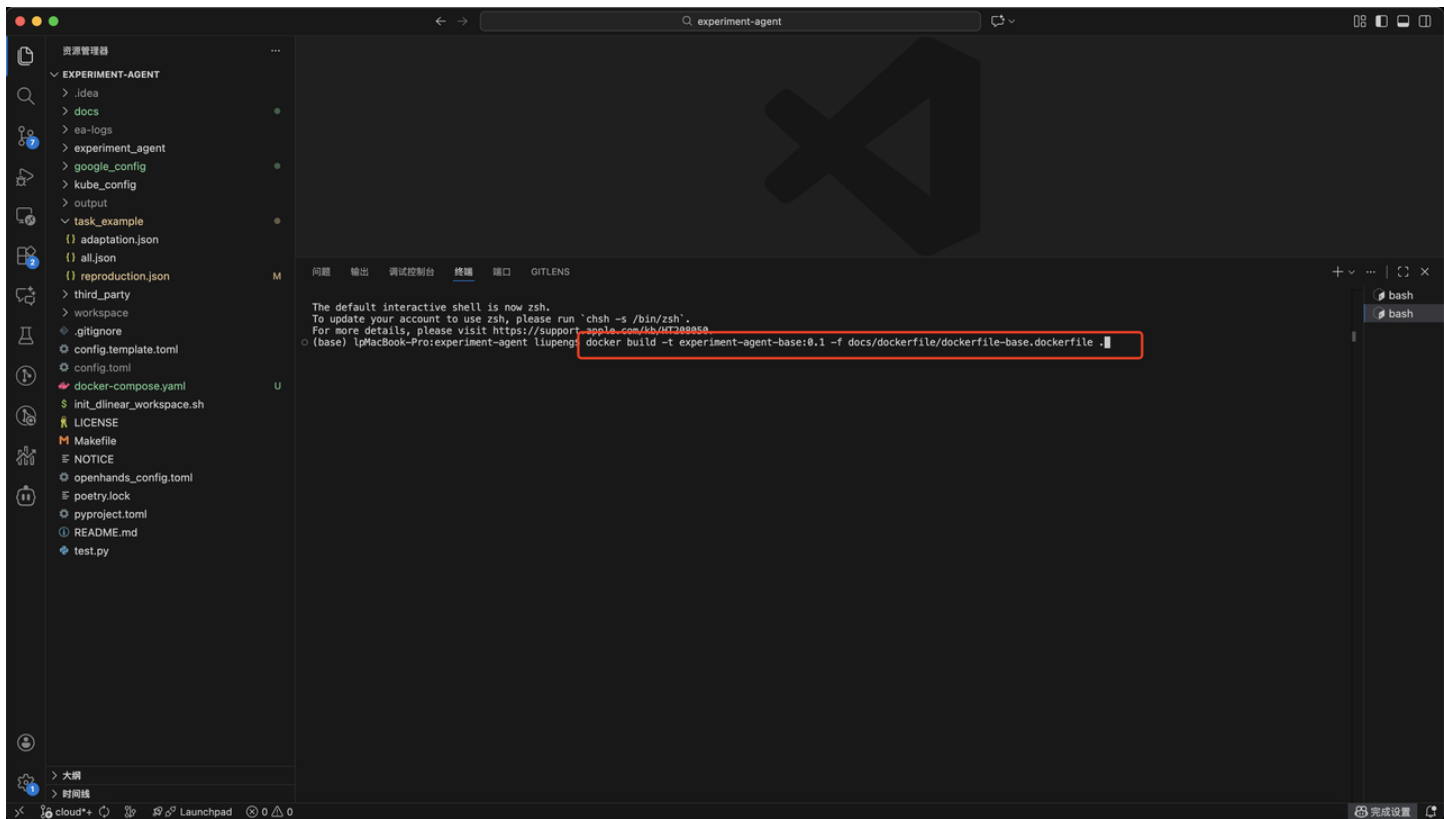
等待命令执行完毕,完成ea依赖安装

4.镜像构建

运行命令构建ea使用的基础镜像

代码块

```
1 docker build -t experiment-agent-base:0.1 -f docs/dockerfile/dockerfile-  
base.dockerfile .
```



配置准备

环境配置



config.toml

1.30KB



将配置文件放到项目根目录下,添加llm配置相关配置

代码块

```
1  [llm]
2  # 模型名称, 例: claude-sonnet-4-5
3  model = ""
4  api_key = ""
5  base_url = ""
6  temperature = 0
7  max_message_chars= 30_000
```

任务相关配置

1. cd 到工程目录下

The image shows a screenshot of an IDE interface. On the left is a file explorer showing the project structure of 'experiment-agent'. The main area on the right is a code editor displaying the 'taion_result.py' file. At the bottom is a terminal window with three tabs: 'Terminal', 'Local (2)', and 'Local (3)'. The 'Local (3)' tab is active, showing a shell prompt and the execution of 'pwd' and 'main'.

Project Structure:

- experiment-agent
 - docs
 - experiment_agent
 - output
 - task_example
 - third_party
 - workspace
 - .gitignore
 - 0_install_docker.sh
 - 1_install_NVIDIA_Container_Toolkit.sh
 - 2_install_conda.sh
 - 3_install_ea.sh
 - config.template.toml
 - config.toml
 - init_dlinear_workspace.sh
 - LICENSE
 - Makefile
 - NOTICE
 - openhands_config.toml
 - poetry.lock
 - pyproject.toml
 - pyproject.toml.bak
 - README.md
 - test.py
- External Libraries
- Scratches and Consoles

Code Editor (taion_result.py):

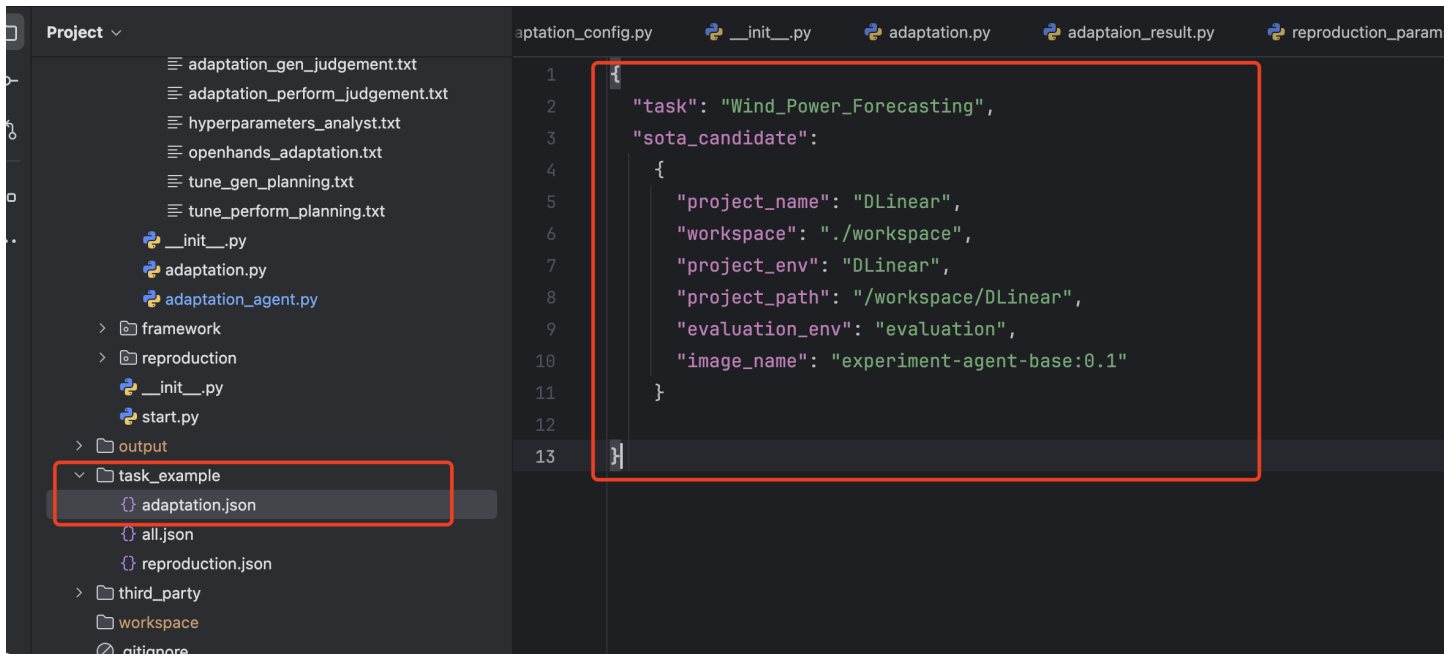
```
1 # agent执行的阶段
2 stage = 'adaptation'
3 # agent执行命令
4 runtime='docker'
5 # 是否开启gpu
6 enable_gpu = True
7 [google_k8s]
8 # gpu个数
9 gpu_count = 1
10
11 [local_docker]
12 # 对应项目、数据集名称
13 workspace = 'workspace'
14 # 配置对应的gpu
15 cuda_visible_devices = '0'
16
17 # 大模型接口配置
18 [llm]
19 # 模型名称，例如：qwen2.5-72b-instruct
20 model = "qwen2.5-72b-instruct"
21 api_key = ""
22 base_url = "http://localhost:8080/v1"
23 temperature = 0.7
24 max_message_count = 10
25
```

Terminal:

```
* cloud
main
(experiment-agent) jcl@jianchunlideMBP experiment-agent % pwd
/Users/jcl/Documents/GitHub/experiment-agent
(experiment-agent) jcl@jianchunlideMBP experiment-agent %
```

2. 修改task_example/adaptation.json文件参数为对应相关工作信息:

根据要求修改相关参数（task参数可以忽略）



项目启动：

代码块

```
1 python experiment_agent/start.py --params-file task_example/adaptation.json
```

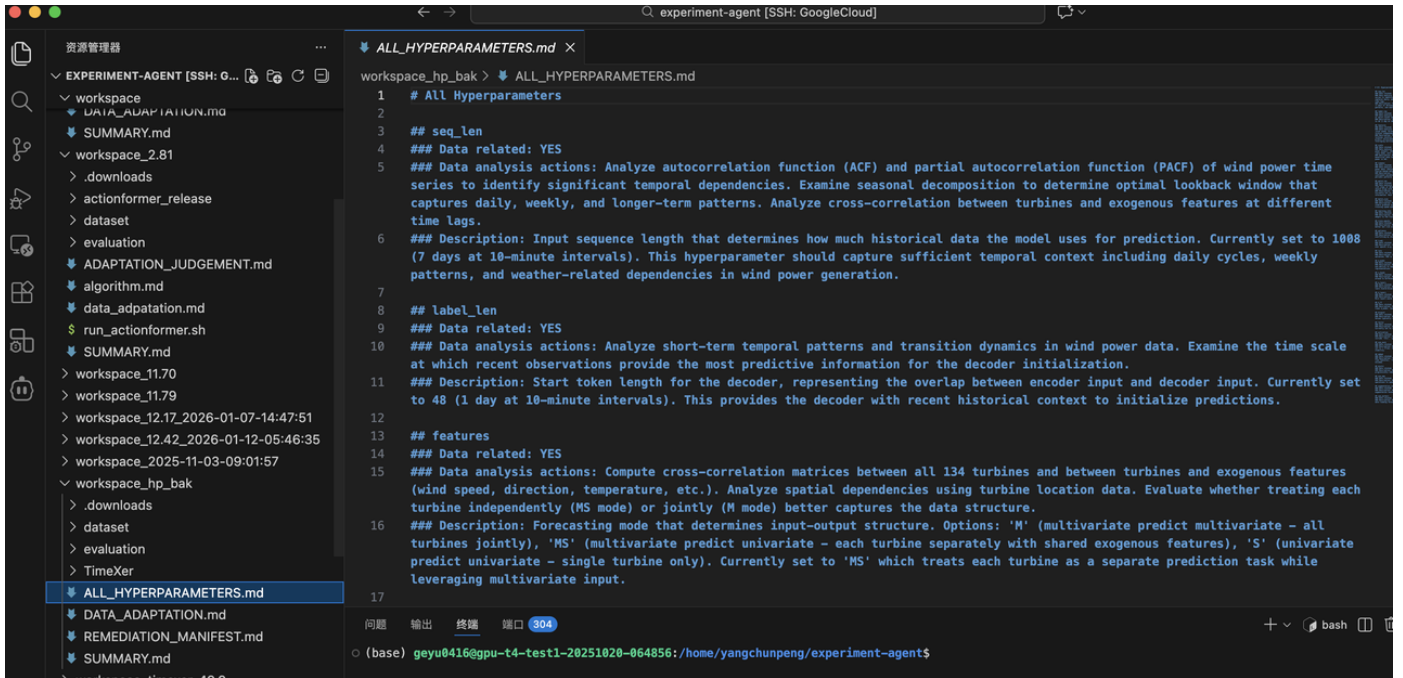
3. 完成后查看日志获取适配是否成功与最终得分情况

EA端到端进行调优

调优：

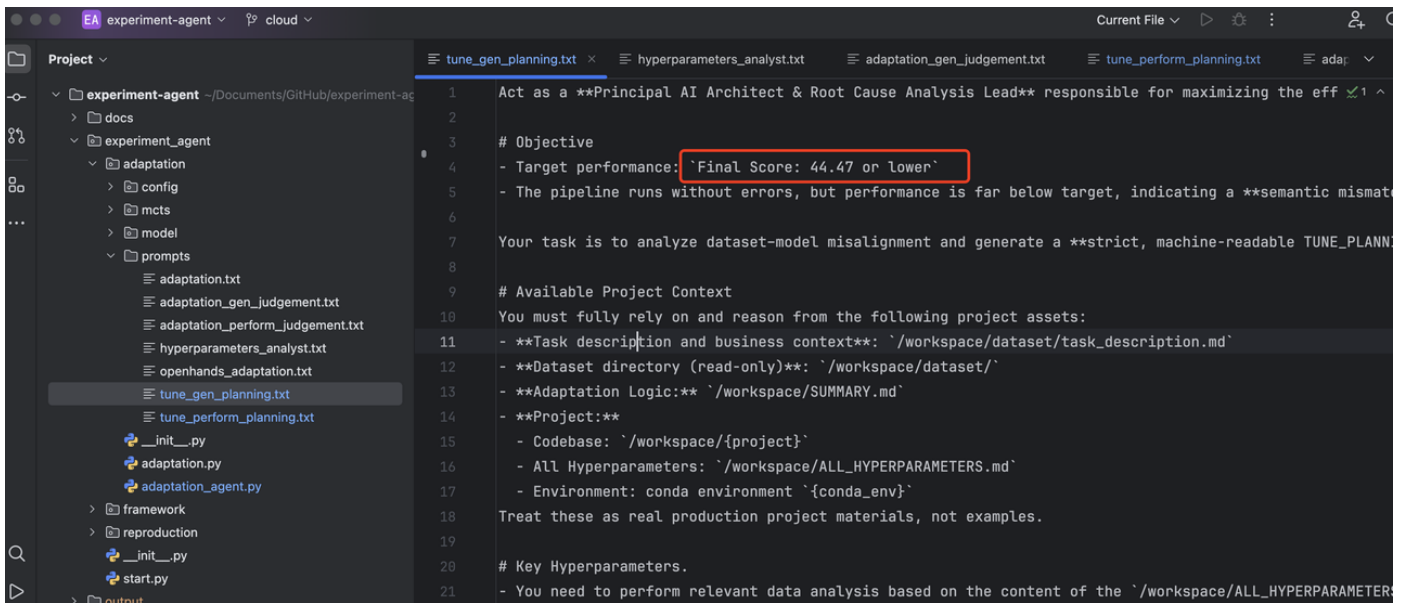
第一步（生成当前算法的超参数分析，该步只需跑一次）

1. 打开**adaptation_agent.py**注释其他prompt加载，只打开**hyperparameters_analyst.txt**并执行生成文件 `/workspace/ALL_HYPERPARAMETERS.md`



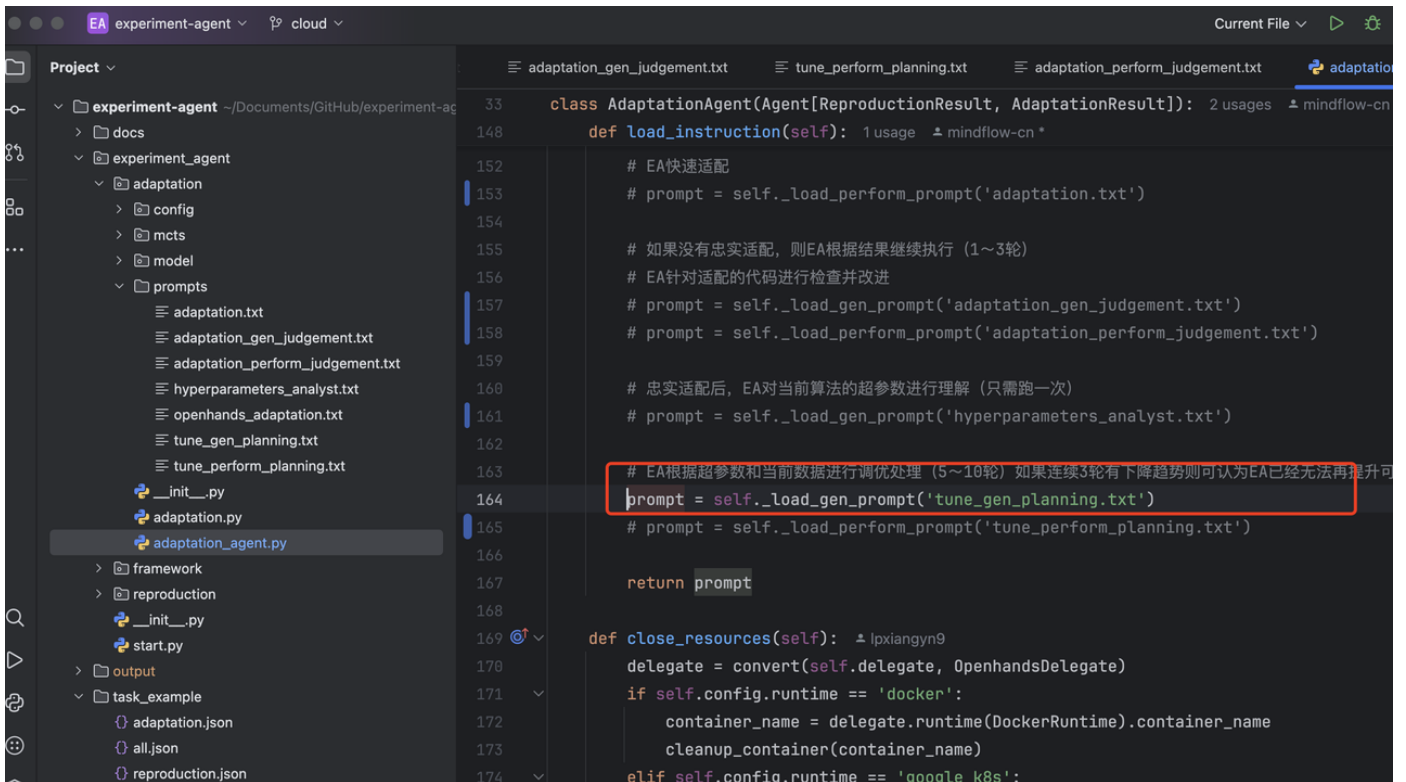
Tip: 该步骤最多进行10轮的调优循环，10轮中取出最优效果即可，如果连续3轮都有下降趋势则可提前停止

1. 打开 `tune_gen_planning.txt` 和 `tune_perform_planning.txt`，添加需要调优后达到的最佳分数（该分数在对应的sow中的比赛可以获取，直接拿比赛第一名的得分作为目标即可）



```
1 Act as a Principal AI Architect & Root Cause Analysis Lead responsible for maximizing the eff
2
3 # Objective
4 - Target performance: 'Final Score: 44.47 or lower'
5 - The pipeline runs without errors, but performance is far below target, indicating a semantic mismatch
6
7 Your task is to analyze dataset-model misalignment and generate a strict, machine-readable TUNE_PLANNING
8
9 # Available Project Context
10 You must fully rely on and reason from the following project assets:
11 - Task description and business context: '/workspace/dataset/task_description.md'
12 - Dataset directory (read-only): '/workspace/dataset/'
13 - Adaptation Logic: '/workspace/SUMMARY.md'
14 - Project:
15   - Codebase: '/workspace/{project}'
16   - All Hyperparameters: '/workspace/ALL_HYPERPARAMETERS.md'
17   - Environment: conda environment '{conda_env}'
18 Treat these as real production project materials, not examples.
19
20 # Key Hyperparameters.
21 - You need to perform relevant data analysis based on the content of the '/workspace/ALL_HYPERPARAMETERS'
```

2. 注释其他prompt加载，使用 `tune_gen_planning.txt` 并执行



```
33 class AdaptationAgent(Agent[ReproductionResult, AdaptationResult]):
34     def load_instruction(self):
35         # EA快速适配
36         # prompt = self._load_perform_prompt('adaptation.txt')
37
38         # 如果没有忠实适配，则EA根据结果继续执行（1~3轮）
39         # EA针对适配的代码进行检查并改进
40         # prompt = self._load_gen_prompt('adaptation_gen_judgement.txt')
41         # prompt = self._load_perform_prompt('adaptation_perform_judgement.txt')
42
43         # 忠实适配后，EA对当前算法的超参数进行理解（只需跑一次）
44         # prompt = self._load_gen_prompt('hyperparameters_analyst.txt')
45
46         # EA根据超参数和当前数据进行调优处理（5~10轮）如果连续3轮有下降趋势则可认为EA已经无法再提升可
47         prompt = self._load_gen_prompt('tune_gen_planning.txt')
48         # prompt = self._load_perform_prompt('tune_perform_planning.txt')
49
50     return prompt
51
52 def close_resources(self):
53     delegate = convert(self.delegate, OpenhandsDelegate)
54     if self.config.runtime == 'docker':
55         container_name = delegate.runtime(DockerRuntime).container_name
56         cleanup_container(container_name)
57     elif self.config.runtime == 'google k8s':
```

3. 注释其他prompt加载，使用 `tune_perform_planning.txt` 并执行

```
33 class AdaptationAgent(Agent[ReproductionResult, AdaptationResult]): 2 usages mindflow-cn +2 +
148 def load_instruction(self): 1 usage mindflow-cn +
152 # EA快速适配
153 # prompt = self._load_perform_prompt('adaptation.txt')
154
155 # 如果没有忠实适配, 则EA根据结果继续执行 (1~3轮)
156 # EA针对适配的代码进行检查并改进
157 # prompt = self._load_gen_prompt('adaptation_gen_judgement.txt')
158 # prompt = self._load_perform_prompt('adaptation_perform_judgement.txt')
159
160 # 忠实适配后, EA对当前算法的超参数进行理解 (只需跑一次)
161 # prompt = self._load_gen_prompt('hyperparameters_analyst.txt')
162
163 # EA根据超参数和当前数据进行调优处理 (5~10轮) 如果连续3轮有下降趋势则可认为EA已经无法再提升可停止
164 # prompt = self._load_gen_prompt('tune_gen_planning.txt')
165 prompt = self._load_perform_prompt('tune_perform_planning.txt')
166
167 return prompt
168
169 def close_resources(self): 1 usage mindflow-cn +
170 delegate = convert(self.delegate, OpenhandsDelegate)
171 if self.config.runtime == 'docker':
172     container_name = delegate.runtime(DockerRuntime).container_name
173     cleanup_container(container_name)
174 elif self.config.runtime == 'google_k8s':
```

4. 查看 /workspace/SUMMARY.md ，调优结果

```
1 # Summary Report
2
3 ## Executive Summary
4
5 I successfully implemented all 6 optimization steps from the REMEDIATION_MANIFEST.md to improve the ActionFormer model's temporal
6 localization precision. The optimization focused on extending training duration, increasing regression loss weight, adjusting center
7 sample radius, reducing label smoothing, and tuning NMS parameters.
8
9 ## Performance Results:
10 - **Previous Performance:** 10.95% Average mAP (baseline after remediation)
11 - **Current Performance:** 12.42% Average mAP (using epoch_015 checkpoint)
12 - **Improvement:** +1.47% absolute improvement (+13.4% relative improvement)
13 - **Target Achievement:** 12.42% / 12.15% = **102.2% of target achieved** ✓
14
15 ## Key Achievement:
16 The optimization successfully exceeded the target performance of 12.15% mAP, achieving 12.42% mAP through
17 careful hyperparameter tuning and extended training.
18
19 ## Applied Optimizations
20
21 All 6 steps from REMEDIATION_MANIFEST.md were implemented exactly as specified:
22
23 ### STEP 1: Extend Training Epochs (CRITICAL)
24 - **File:** `workspace/actionformer_release/configs/perception_test.yaml` (line 55)
25 - **Change:** `epochs: 30 -> 45`
26 - **Rationale:** Training loss at epoch 32 was 0.24 and still decreasing. Extended training allows better convergence.
27 - **Impact:** Enabled the model to reach better generalization at earlier epochs (15-25)
28 - **Expected improvement:** +0.5-1.0% mAP
29
30 ### STEP 2: Increase Regression Loss Weight (CRITICAL)
```